IT Project Management

Lecture 1:
Introduction to Project Management

Dr. Hung-Nghiep Tran
nghiepth@uit.edu.vn

University of Information Technology, VNU-HCM, Vietnam
2025-2026

About this lecture

- You will learn:
  - Definition and examples of a project
  - Attributes and categorization of projects
  - Definition and constraints of project management
  - Key elements of the project management framework

Why project management?

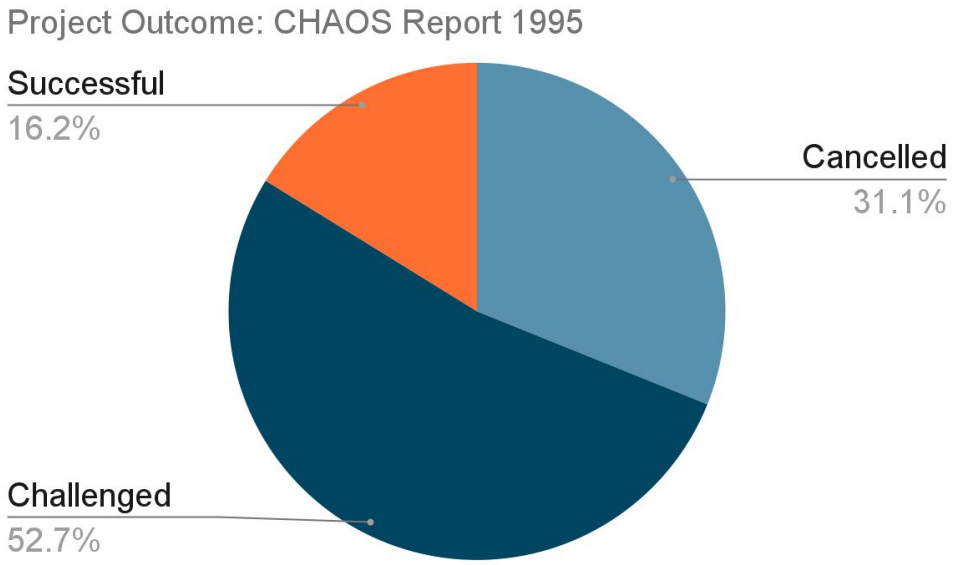
Why Project Management?

- Worldwide IT spending: $3.5 trillion in 2017
  - Increasing ~2.4% yearly
- The number of PM jobs ~66 million in 2017
  - ~87.7 million jobs by 2027*

*Project Management Institute (PMI) Report

Why Project Management?

IT Projects have a **terrible track record**:



Why Project Management?

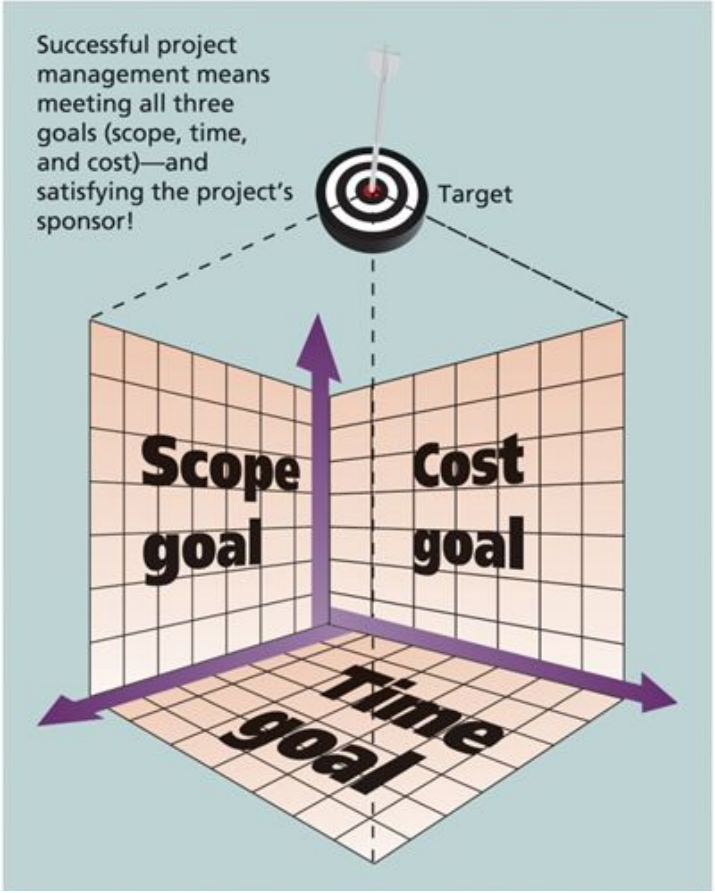
IT Projects have a **terrible track record**:

Only 2.5% of **corporations** consistently **meet their targets for scope, time, and cost goals***

* PricewaterhouseCoopers study

Project Success

- Project success:
  - The project **met scope, time, and cost goals**
  - The project **satisfied the customer/sponsor**
  - The results of the project **met its main objective**

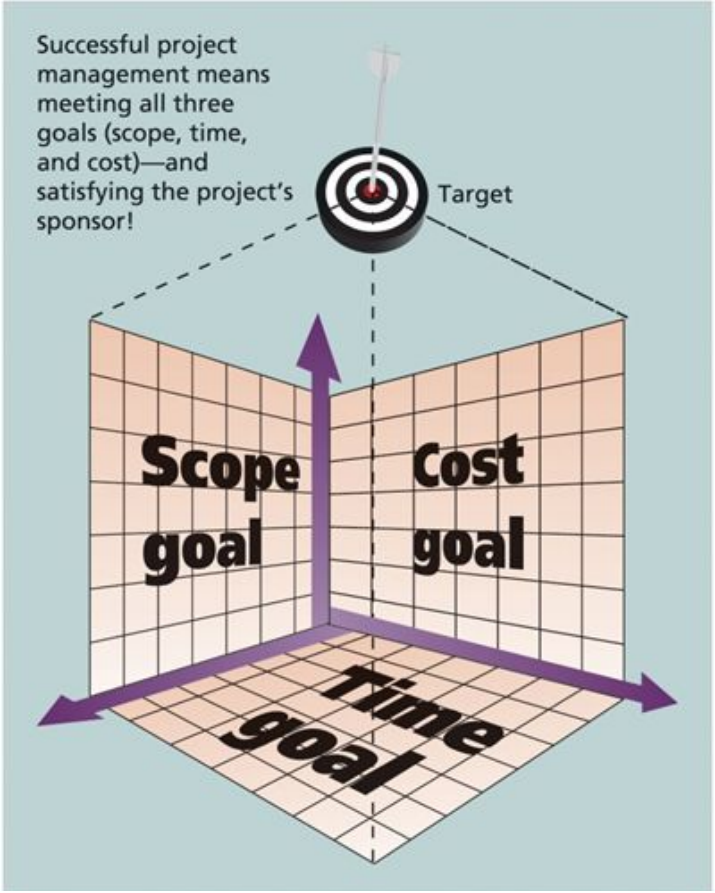


Project Success

- Top reasons for <u>successful</u> projects (from surveys):
  - Adequate funding
  - Good staff expertise
  - Engagement from all stakeholders

Project Failure

- ***Cancelled project:*** terminated before it achieves its purpose
- ***Challenged project:*** done but significantly breaks either scope, time, or cost goals

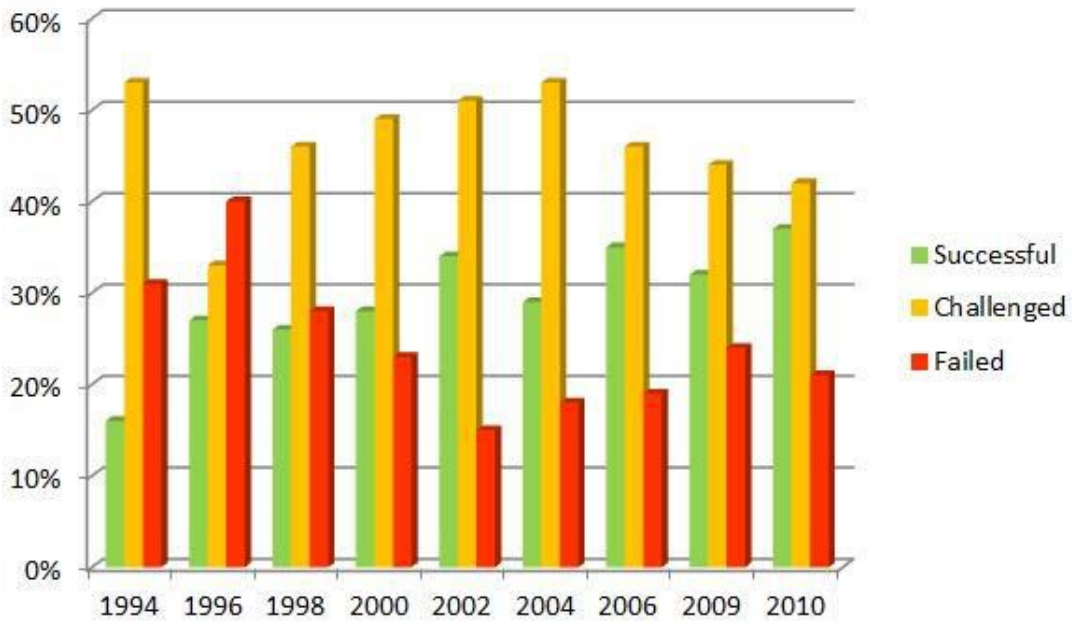


Project Failure

- Top reasons for <u>failed</u> projects (from surveys):
  - **Bad management**
  - Lack of information
  - Ambiguous objectives

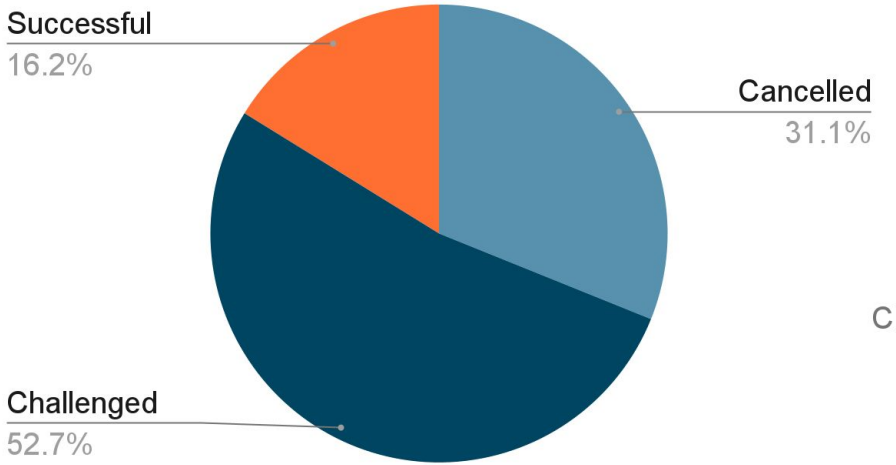
Why Project Management?

Active project management has improved the results over the years:

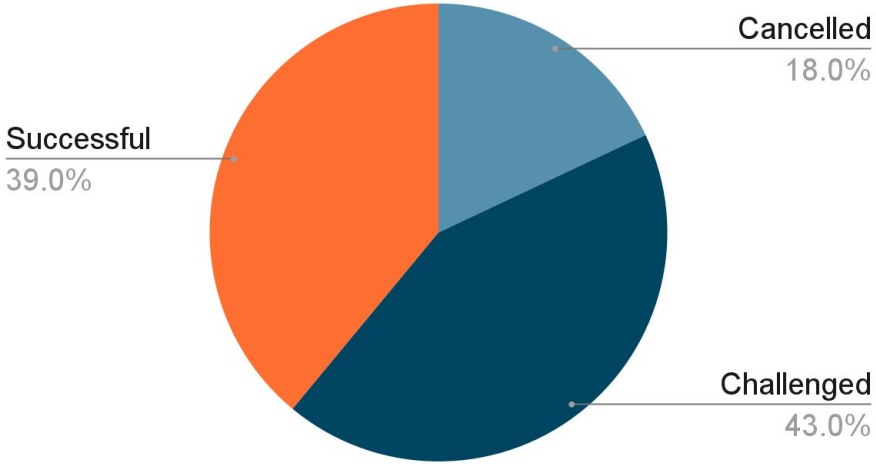


Why Project Management?

Project Outcome: CHAOS Report 1995



CHAOS Report 2012



Advantages of active project management

- Better control of financial, physical, and human resources
- Improved customer relations
- Shorter development times
- Lower costs and improved productivity
- Higher quality and increased reliability
- Higher profit margins
- Better internal coordination
- Positive impact on meeting strategic goals
- Higher worker morale

What is a project?

What is a project?

PMBOK® Guide, Sixth Edition, 2017:

- A project is “a **temporary** endeavor undertaken to create a **unique** product, service, or result”

  - **A project ends** when its objectives have been reached or the project has been terminated

What is a project?

Other definitions:

- A project is a **sequence of tasks** carried out to achieve a **defined objective** under constraints of **scope**, **time**, and **budget**

- A project is a **collection of related activities** aimed at completing a **specific product** within a predetermined **cost and timeframe**

What is a project?

Example

- A company develops a full self-driving car
  - Team: engineers, scientists, etc.
  - Output: a full self-driving system
  - Timeframe: years (Tesla: since 2016-Now, not finished yet)
  - Cost: billions dollars (Tesla: 10+ billions)

What is a project?

Examples

- A government group develops a system to track vaccinations
- A bank acquires other financial institutions and consolidate its systems
- A team of students creates an AI-powered web application

What is a project?

Attributes

- A project
  - **has a specific purpose**
  - **is temporary with start and end dates**
  - drives change and enable value creation
  - is developed using progressive elaboration
  - requires resources, often from various areas
  - should have a primary customer or sponsor
    - (customers pay when using the output of the project)
    - (sponsors pay when funding the work of project)
  - involves uncertainty

What is a project?

Categorization

- By project domain: IT projects, construction projects, etc.
- By project size: small, medium, large projects
- By scope, cost, timeframe, etc.

What is an IT project?

- An IT project
  - is a project
    - with all attributes of a general project
  - and is related to IT (hardware, software, systems, etc.)

- An IT project has some **special attributes**:
  - Involves more uncertainties
  - Dynamic, technology-driven
  - Iterative structure, with frequent feedback and changes
  - Uses one of the software development processes

What is project management?

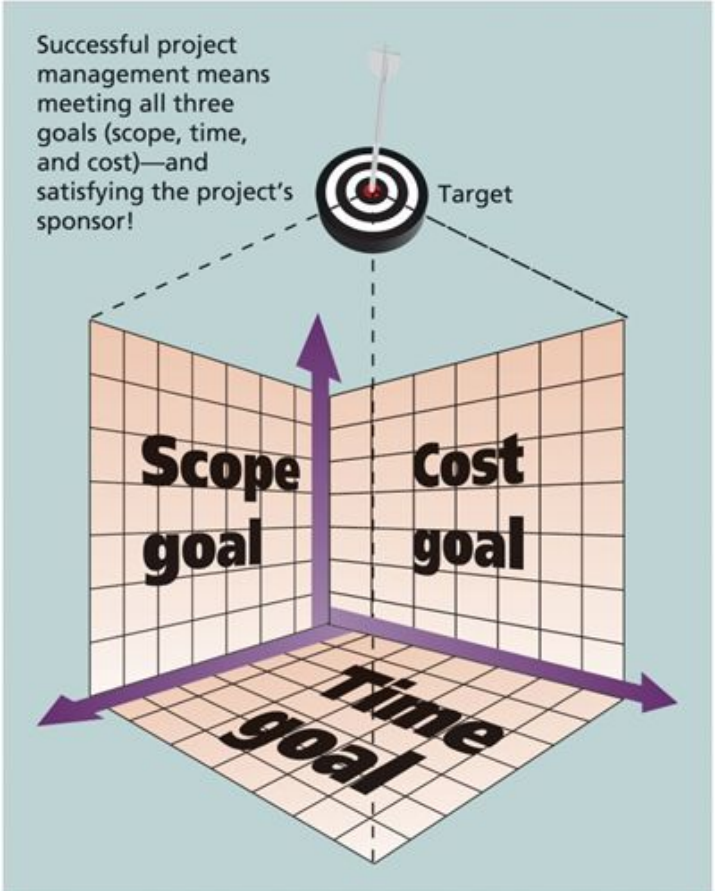
What is Project Management?

PMBOK® Guide, Sixth Edition, 2017:

- Project management is “the application of **knowledge, skills, tools and techniques** to **project activities** to meet project requirements”

What is Project Management?

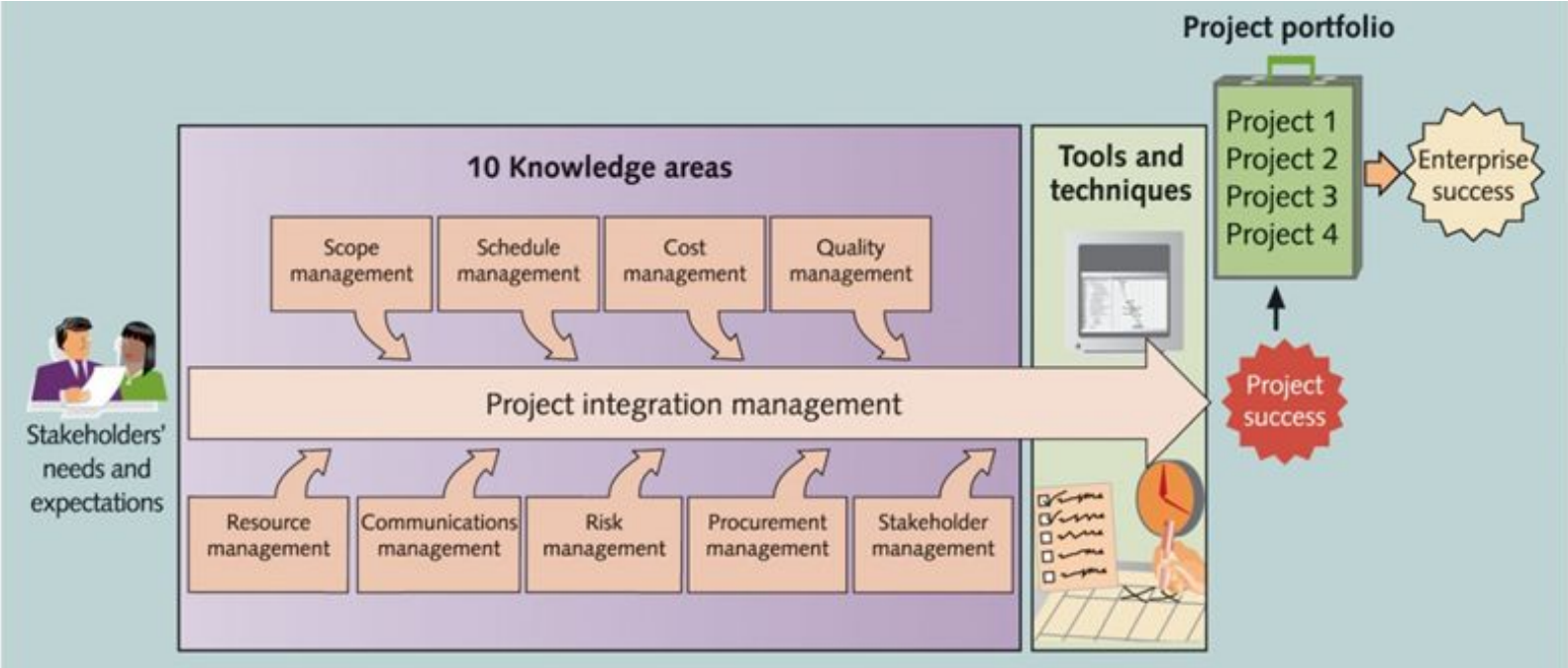
- **Project managers** strive to help the project succeed:
  - meet the **triple constraint (scope, time, and cost goals)**
  - and also meet the needs and expectations of project stakeholders



What is Project Management?

10 knowledge areas

Knowledge areas describe the **key competencies** that PMs must develop



What is Project Management?

Tools & Techniques

- Project management tools and techniques assist project managers and their teams in various aspects of project management
- For example:
  - **Scope:** Project charter, scope statement, and WBS (work breakdown structure)
  - **Time:** Gantt charts, network diagrams, critical path analysis, critical chain scheduling
  - **Cost:** Cost estimates and earned value management

What is Project Management?

Stakeholders

- Stakeholders: the **people involved** in or **affected** by project activities
- Stakeholders include:
  - the project sponsor
  - banks and other financial institutions
  - the project manager
  - the project team
  - support staff
  - suppliers
  - ***opponents*** to the project

From a Broader View:

Program and Project Portfolio Management

- Two important concepts that help align projects to enterprise goals:
  - **Programs**
  - Project **portfolio** management

From a Broader View:
Programs

- A **program** is “a group of related projects managed in a **coordinated manner** to **obtain benefits** and control **not** available from managing them **individually**” – PMBOK® Guide Sixth Edition, 2017

  Examples: programs in the **IT field** include **infrastructure**, **applications development**, and **user support**

- A **program manager** provides **leadership** and **direction** for the project managers heading the **projects within the program**

From a Broader View:
Portfolio

- As part of project portfolio management, organizations group and **manage projects and programs as a portfolio of investments** that contribute to the entire enterprise's success

- **Portfolio managers** help their organizations make wise investment decisions by helping to **select and analyze projects from a strategic perspective**

From a Broader View:

Portfolio

Project Manager

Roles

- Project managers must work closely with the other stakeholders on a project, especially the sponsor and project team
- They are also more effective if they are familiar with
  - the 10 project management **knowledge areas**
  - the **various tools and techniques** related to project management

Project Manager

Job Descriptions

- Job descriptions vary, but most include responsibilities like **planning, scheduling, coordinating, and working with people** to achieve project goals
- Project management is a skill needed in every major IT field, from database administrator to network specialist to technical writer

Project Manager

Advices for young professionals about career path

- A few questions to ask yourself:
  - Do you get frustrated by bad bosses? Do you think you could do a better job?
  - Are you interested in understanding the big picture of how organizations work and how your individual work or your project fits in?
  - Have you had other leadership roles, such as being a team captain, president of a club, or entrepreneur of a small business? Did you enjoy it? Did others think you did a good job?
  - Are you good at mentoring others? Do people ask you for help in developing their skills or your advice on what to do?

Project Manager

Suggested skills

- The Project Management Body of Knowledge
- Application area knowledge, standards, and regulations
- Project environment knowledge
- General management knowledge and skills
- Soft skills or human relations skills

Project Manager

Suggested skills

- Six traits of highly effective project managers as follows:
  - Be a strategic business partner
  - Encourage and recognize valuable contributions
  - Respect and motivate stakeholders
  - Be fully vested in success
  - Stress integrity and accountability
  - Work in the gray/Be able to deal with ambiguity

Project Manager

The PMI's talent triangle

- **Technical** skills: coding, technical reading/writing, etc.
- **Business** skills: management, accounting, etc.
- **People** skills: leadership, communication, etc.

Project Manager

Careers in IT

- Even if you choose to stay in a technical role, **you still need project management knowledge and skills** to help your team and organization

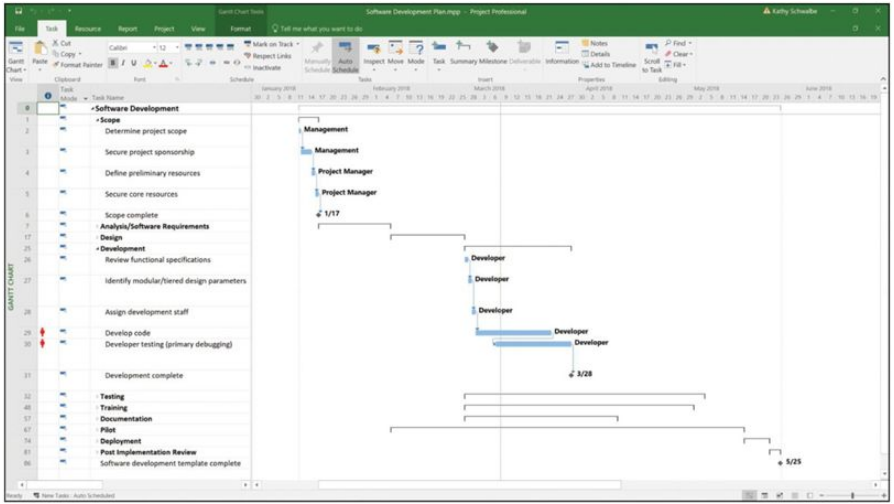
1. Full-stack software development
2. Project management
3. Cyber-security
4. Networking
5. User experience/user interface (UX/UI) design
6. Quality assurance (QA)/testing
7. Cloud engineering
8. Big data
9. Machine learning/artificial intelligence
10. DevOps

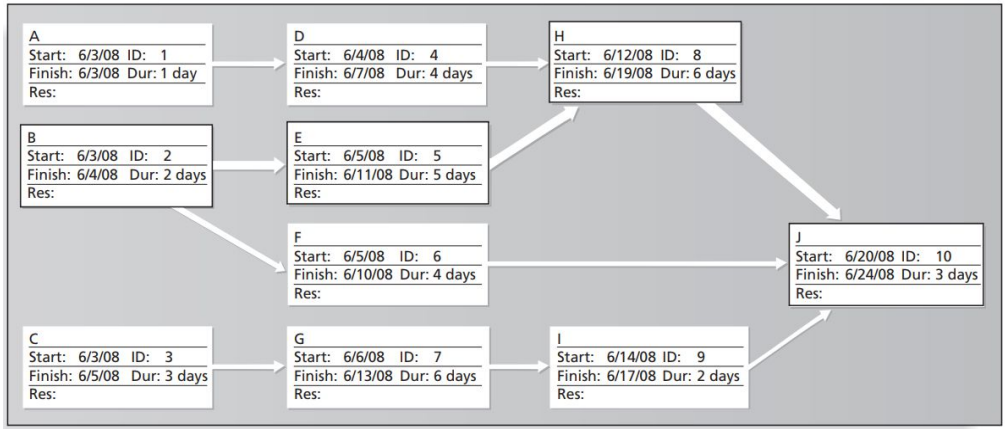
2017 survey: “10 hot tech skills”

The Project Management Profession

History

- In 1917, Henry Gantt developed the famous **Gantt chart** for scheduling work.
- In 1958, the U.S. Navy polaris missile/submarine project first used **network diagrams**.

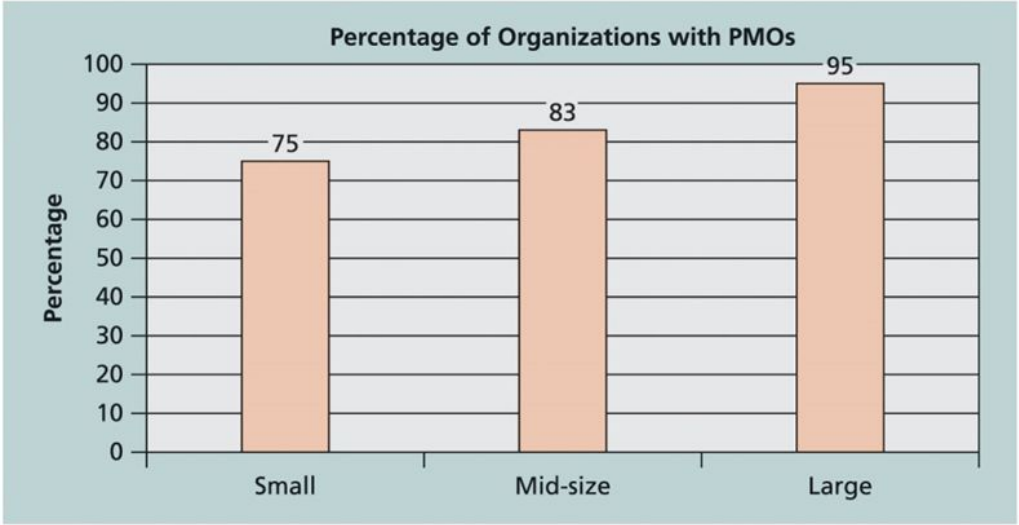




The Project Management Profession

History

- In the 1990s, many companies began creating **Project Management Offices** (PMO) to help them handle the increasing number and complexity of projects



Source: PM Solutions, "The State of the Project Management Office (PMO) 2016," 2016.

The Project Management Profession

Emerging Global Issues

- Several global dynamics are forcing organizations to rethink their practices:
  - Talent development for project and program managers is a top concern
  - Basic project management techniques are core competencies
  - Organizations want to use more agile approaches to project management
  - Benefits realization of projects is a key metric

The Project Management Profession

Project Management Software

- There are hundreds of different products to assist in performing project management
- Three main categories of tools:
  - Low-end tools: Handle single or smaller projects well, cost under $200 per user
  - Midrange tools: Handle multiple projects and users, cost $200-$1,000 per user, Microsoft Project is still one of the most popular
  - High-end tools: Also called enterprise project management software, often licensed on a per-user basis
- Several free or open-source tools are also available

The Project Management Profession

Project Management Software

- Open sources:
  - OpenProject
  - ProjectLibre
  - ODOO
  - Wekan
  - ZenTao
  - …

The 10 Top-Rated Free Project Management Software Options

	Free users	Free projects/ project workspaces	Free storage	Cost to upgrade	Work management	Project planning (Define scope, timeline, resources)
Airtable	1 user, Unlimited collaborators	Unlimited bases, 1200 records per base	2GB/base	$12/user/ month	✓	
avaza	1 user, Unlimited collaborators	5 projects, unlimited archived projects	100MB	$9.95/user/ month	✓	✓
backlog	10	1	100MB, one 5MB attach. per issue	$20/month (up to 30 users)	✓	
clickup	Unlimited	Unlimited	100MB	$9/user/ month	✓	
Flowlu	2	1	1GB	$39/month (up to 5 users)	✓	✓
freedcamp	Unlimited	Unlimited	Unlimited, 10MB upload	$2.49/user/ month	✓	
meistertask	Unlimited	Unlimited	Unlimited, 20MB attachment	$8.25/user/ month	✓	
paymo	1	Unlimited	1GB	$11.95/user /month	✓	✓
STREAMTIME	Unlimited	5 active jobs	50 archived jobs	$15/user/ month	✓	✓
teamwork Projects	5	2	100MB	$11.25/user/ month (min. of 5 users)	✓	✓

Capterra

The Project Management Institute (PMI)

- The Project Management Institute (PMI) is an international professional society for project managers founded in 1969
- PMI has continued to attract and retain members, reporting more than 500,000 members worldwide by late 2017
- There are communities of practices in many areas, like information systems, financial services, and health care
- Provide **guidelines, examinations, and certifications** for professional project managers around the world

The Project Management Institute (PMI)

Certification

- PMI provides Project Management Professional (PMP®) certification for those with:
  - Documented sufficient project experience
  - Agreeing to follow a code of ethics
  - Having passed the PMP® exam

Source: Data points from Project Management Institute, PMI Today® (annual issues).

Summary

- You have learned:
  - Definition and examples of a project
  - Attributes and categorization of projects
  - Definition and constraints of project management
  - Key elements of the project management framework, including:
    - project stakeholders, the project management knowledge areas, and common tools and techniques

Thank you for listening

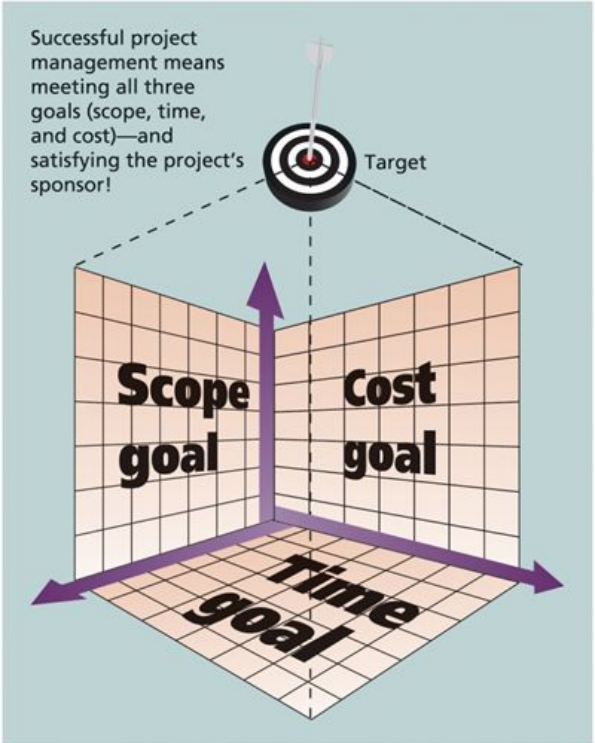
Q & A

Appendix

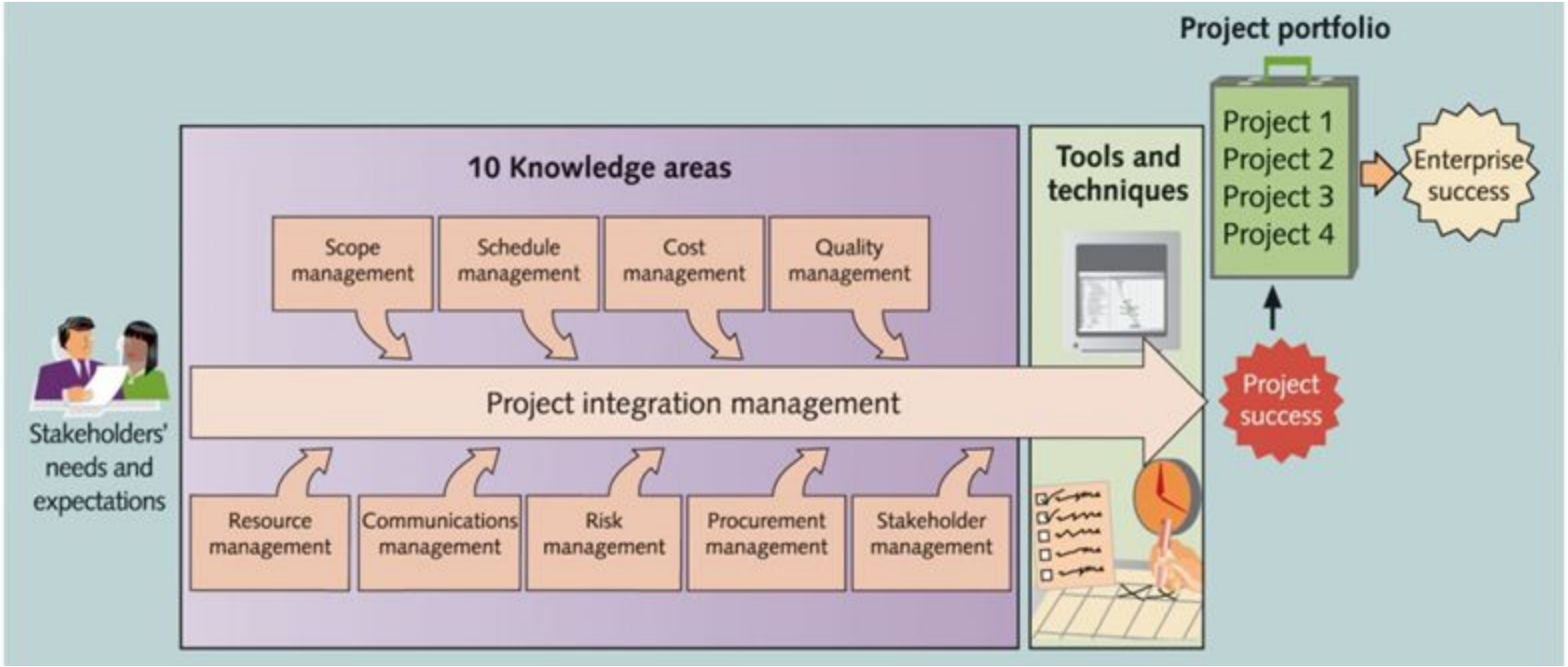
Recap & Preview

Project Success

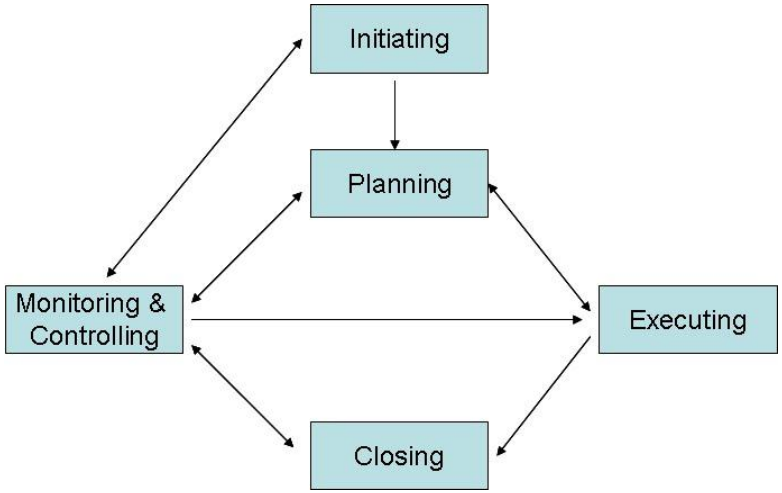
- Project success:
  - The project **met scope, time, and cost goals**
  - The project **satisfied the customer/sponsor**
  - The results of the project **met its main objective**

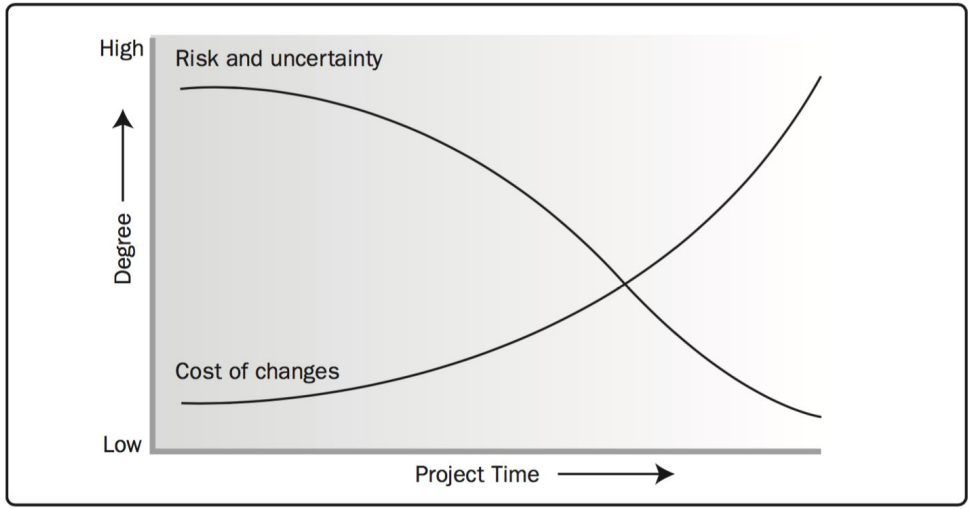


10 knowledge areas

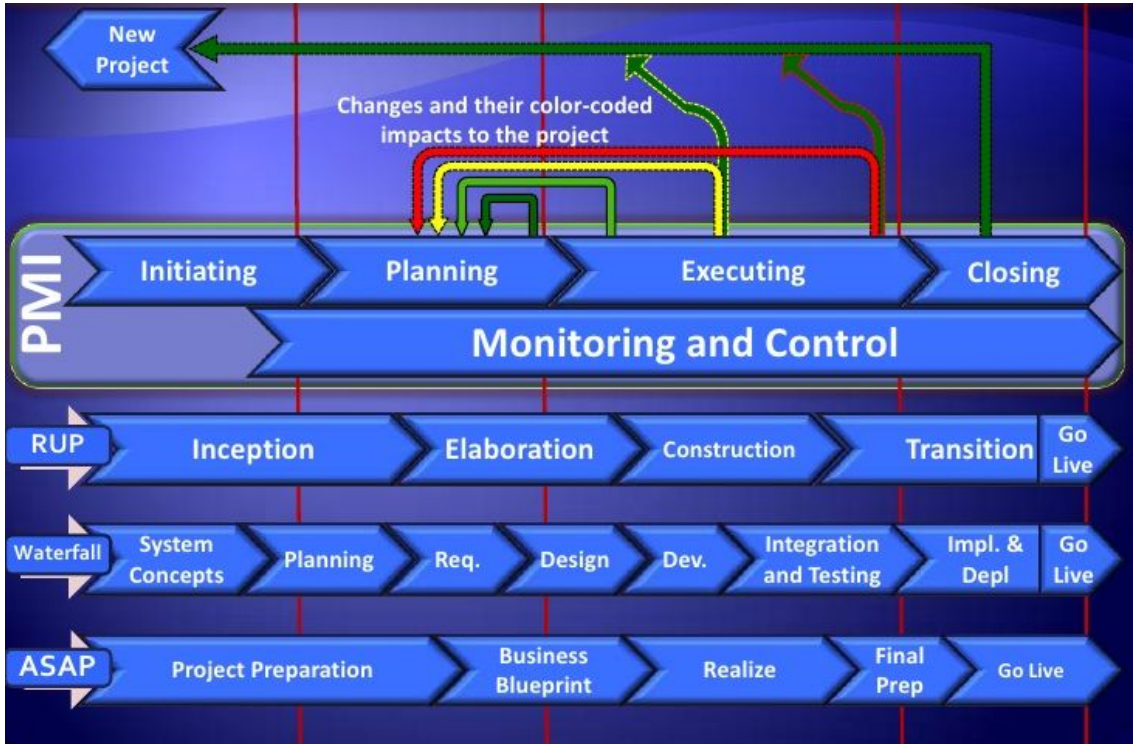


Project Life Cycle





SDLC: System Development Life Cycle

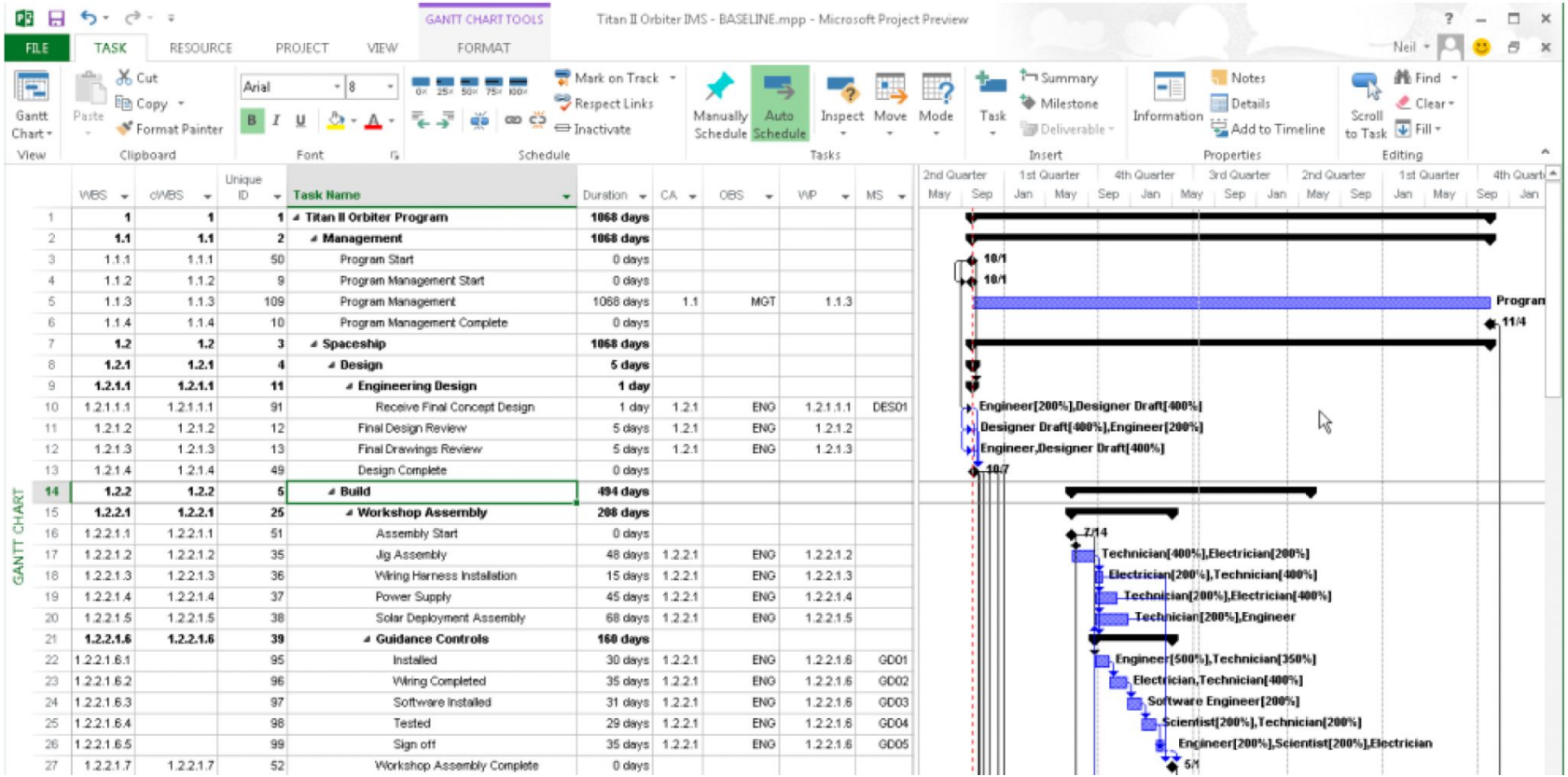


The Process Groups and Knowledge Areas

Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
Project Integration Management	Develop project charter	Develop project management plan	Direct and manage project execution	Monitor and control project work, Perform integrated change control	Close project or phase
Project Scope Management		Collect requirements, Define scope, Create WBS		Verify scope, Control scope	
Project Time Management		Define activities, Sequence activities,		Control schedule	

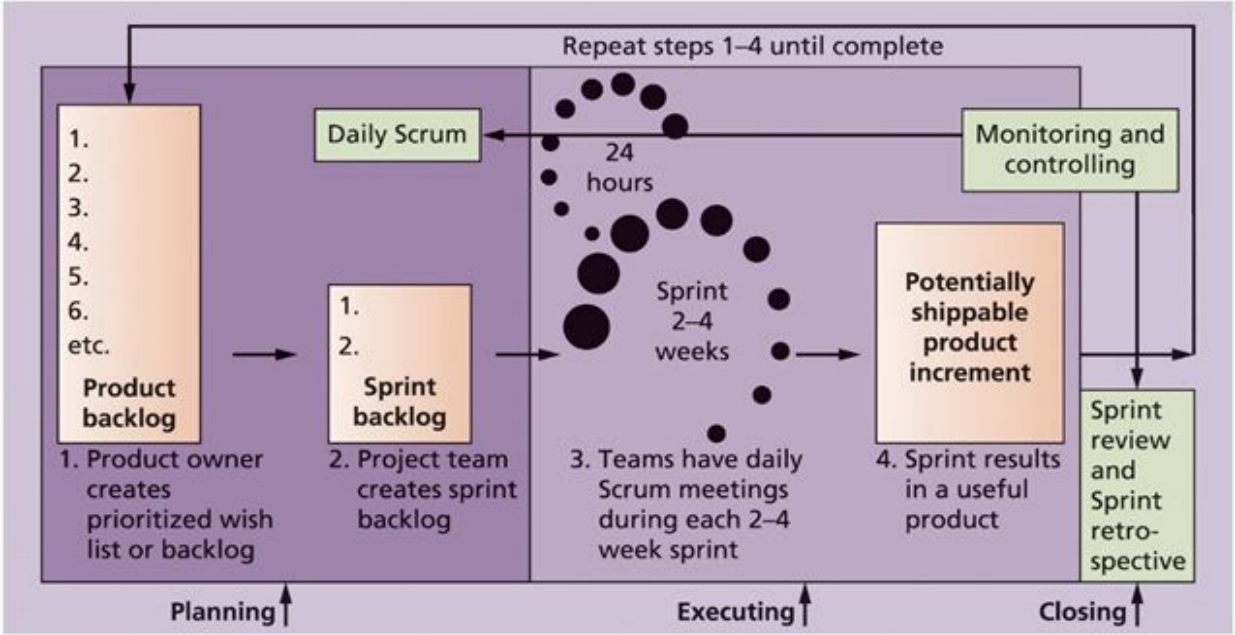
Knowledge Area	Project Management Process Groups				
	Initiating	Planning	Executing	Monitoring and Controlling	Closing
Project Time Management (continued)		Estimate activity resources, Estimate activity durations, Develop schedule			
Project Cost Management		Estimate costs, Determine budget		Control costs	
Project Quality Management		Plan quality	Perform quality assurance	Perform quality control	
Project Human Resource Management		Develop human resource plan	Acquire project team, Develop project team, Manage project team		
Project Communi-cations Management	Identify stake-holders	Plan communi-cations	Distribute information, Manage stakeholders expectations	Report performance	
Project Risk Management		Plan risk man-agement, Identi-fy risks, Perform qualitative risk analysis, Perform quantitative risk analysis, Plan risk responses		Monitor and control risks	
Project Procurement Management		Plan procurements	Conduct procurements	Administer procurements	Close procurements

Gantt Chart


GANTT CHART TOOLS
Titan II Orbiter IMS - BASELINE.mpp - Microsoft Project Preview
FILE
TASK
RESOURCE
PROJECT
VIEW
FORMAT
Neil
Gantt Chart
View
Cut
Copy
Paste
Format Painter
Clipboard
Arial
8
Font
Mark on Track
Respect Links
Inactivate
Schedule
Manually Schedule
Auto Schedule
Inspect
Move
Mode
Task
Summary
Milestone
Deliverable
Insert
Information
Notes
Details
Add to Timeline
Properties
Scroll to Task
Find
Clear
Fill
Editing
GANTT CHART
WBS
oWBS
Unique ID
Task Name
Duration
CA
OBS
WP
MS
1 1 1 1 Titan II Orbiter Program 1068 days
2 1.1 1.1 2 Management 1068 days
3 1.1.1 1.1.1 50 Program Start 0 days
4 1.1.2 1.1.2 9 Program Management Start 0 days
5 1.1.3 1.1.3 109 Program Management 1068 days 1.1 MGT 1.1.3
6 1.1.4 1.1.4 10 Program Management Complete 0 days
7 1.2 1.2 3 Spaceship 1068 days
8 1.2.1 1.2.1 4 Design 5 days
9 1.2.1.1 1.2.1.1 11 Engineering Design 1 day
10 1.2.1.1.1 1.2.1.1.1 91 Receive Final Concept Design 1 day 1.2.1 ENG 1.2.1.1.1 DES01
11 1.2.1.2 1.2.1.2 12 Final Design Review 5 days 1.2.1 ENG 1.2.1.2
12 1.2.1.3 1.2.1.3 13 Final Drawings Review 5 days 1.2.1 ENG 1.2.1.3
13 1.2.1.4 1.2.1.4 49 Design Complete 0 days
14 1.2.2 1.2.2 5 Build 494 days
15 1.2.2.1 1.2.2.1 25 Workshop Assembly 208 days
16 1.2.2.1.1 1.2.2.1.1 51 Assembly Start 0 days
17 1.2.2.1.2 1.2.2.1.2 35 Jig Assembly 48 days 1.2.2.1 ENG 1.2.2.1.2
18 1.2.2.1.3 1.2.2.1.3 36 Wiring Harness Installation 15 days 1.2.2.1 ENG 1.2.2.1.3
19 1.2.2.1.4 1.2.2.1.4 37 Power Supply 45 days 1.2.2.1 ENG 1.2.2.1.4
20 1.2.2.1.5 1.2.2.1.5 38 Solar Deployment Assembly 68 days 1.2.2.1 ENG 1.2.2.1.5
21 1.2.2.1.6 1.2.2.1.6 39 Guidance Controls 160 days
22 1.2.2.1.6.1 95 Installed 30 days 1.2.2.1 ENG 1.2.2.1.6 GD01
23 1.2.2.1.6.2 96 Wiring Completed 35 days 1.2.2.1 ENG 1.2.2.1.6 GD02
24 1.2.2.1.6.3 97 Software Installed 31 days 1.2.2.1 ENG 1.2.2.1.6 GD03
25 1.2.2.1.6.4 98 Tested 29 days 1.2.2.1 ENG 1.2.2.1.6 GD04
26 1.2.2.1.6.5 99 Sign off 35 days 1.2.2.1 ENG 1.2.2.1.6 GD05
27 1.2.2.1.7 1.2.2.1.7 52 Workshop Assembly Complete 0 days
2nd Quarter
1st Quarter
4th Quarter
3rd Quarter
2nd Quarter
1st Quarter
4th Quarter
May Sep Jan May Sep Jan May Sep Jan May Sep Jan May Sep Jan
10/1
10/1
Program
11/4
Engineer[200%],Designer Draft[400%]
Designer Draft[400%],Engineer[200%]
Engineer,Designer Draft[400%]
10/7
7/14
Technician[400%],Electrician[200%]
Electrician[200%],Technician[400%]
Technician[200%],Electrician[400%]
Technician[200%],Engineer
Engineer[500%],Technician[350%]
Electrician,Technician[400%]
Software Engineer[200%]
Scientist[200%],Technician[200%]
Engineer[200%],Scientist[200%],Electrician
5/1

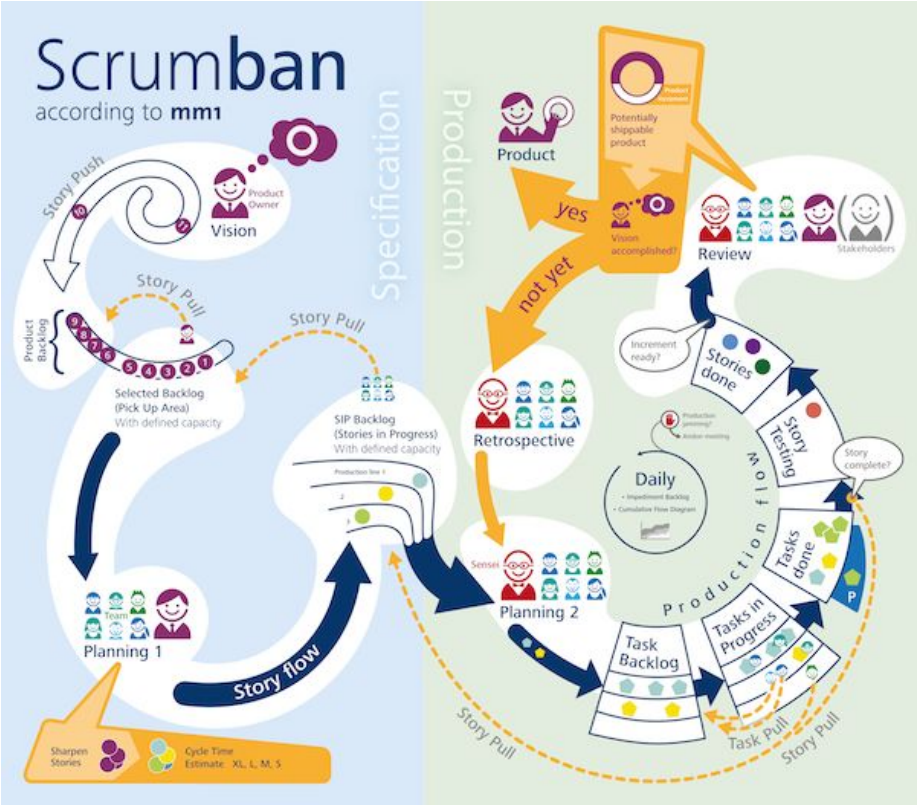
Case Study: Agile Approach

Scrum Roles, Artifacts, and Ceremonies



Scrum framework and the process groups

Agile & Scrum


Scrumban
according to mm1
Specification
Production
Story Push
Product Owner
Vision
Story Pull
Product Backlog
Selected Backlog
(Pick Up Area)
With defined capacity
Story Pull
SIP Backlog
(Stories in Progress)
With defined capacity
Planning 1
Story flow
Sharpen Stories
Cycle Time Estimate XL, L, M, S
Product
Potentially shippable product
yes
not yet
Vision accomplished?
Review
Stakeholders
Increment ready?
Stories done
Story Testing
Story complete?
Retrospective
Daily
Tasks done
Planning 2
Tasks in Progress
Task Backlog
Production flow
Story Pull
Task Pull
Story Pull

A “horror” story
about software development and project management

- The programmer delivers a piece of code, convinced that it has **no bugs**.
- The quality assurance team tests the product and finds **20** bugs.
- The programmer **fixes 10 bugs** and emails the QA team, insisting that the remaining 10 "issues" are not bugs.
- The QA team replies, stating that **5 out of the 10** bug fixes do not work, and attaches a list of **20 new bugs**.
- Meanwhile, the marketing team announces that the launch date, in a few weeks, has been publicly advertised.
- The director calls the project team to check on the progress and **boosts their morale** (promising reward, but mainly threatening punishment!) → Alas! The product finishes before the deadline and gets released.
- **Within a week**, users report **120 bugs**.
- Meanwhile, the lead programmer has applied for a “medical” leave.
- A **"rescue team"** of veteran developers is **urgently assembled**. After a week of **intense work**, they manage to **fix all 120 bugs** and create no new bugs.
- However, users report another **300 bugs** while they are working. And another **700 more bugs** in the next week.
- The director sits quietly at his desk, gives up, and decides to hire **a new programmer** to **rebuild from scratch**.
- The **new programmer** delivers a new piece of code, convinced that it has **no bugs**.